

Lecture 2 - Reconstruction Attacks

1. [Garfinkel-Abowd-Martindale]'19
2. [Pinur-Nissim'03]
3. [Cohen-Nissim'20] - DN'03 made practical

1. Census Attack [Garfinkel - Abowd - Martindale '19]

- census gathers data from population
(age, sex, race)

male female black white

"microdata" $\approx$ raw data

Legal restrictions prevent disclosures

Statistic	Group	Age		
		Count	Median	Mean
1A	Total Population	7	30	38
2A	Female	4	30	33.5
2B	Male	3	30	44
2C	Black or African American	4	51	48.5
2D	White	3	24	24
3A	Single Adults	(D)	(D)	(D)
3B	Married Adults	4	51	54
4A	Black or African American Female	3	36	36.7
4B	Black or African American Male	(D)	(D)	(D)
4C	White Male	(D)	(D)	(D)
4D	White Female	(D)	(D)	(D)
5A	Persons Under 5 Years	(D)	(D)	(D)
5B	Persons Under 18 Years	(D)	(D)	(D)
5C	Persons 64 Years or Over	(D)	(D)	(D)

Note: Married persons must be 15 or over

- redundancies + info
→ Constraints

Solving constraints
→ reconstruction

$A \leq B \leq C$ - Male Ages

$0 \leq A, B, C \leq 125$
Whole numbers

Median = 30 $\rightarrow B = 30$

Mean = 44 $\rightarrow \frac{1}{3}(A+B+C) = 44$

$\Rightarrow \frac{1}{3}(A+C) = 34$

Only 31 sets of A, B, C

$\Rightarrow 300,000$ possible ages
 A, B, C

300k $\rightarrow \approx 30$

Constraints reduced
options a lot

A	B	C	A	B	C	A	B	C
1	30	101	11	30	91	21	30	81
2	30	100	12	30	90	22	30	80
3	30	99	13	30	89	23	30	79
4	30	98	14	30	88	24	30	78
5	30	97	15	30	87	25	30	77
6	30	96	16	30	86	26	30	76
7	30	95	17	30	85	27	30	75
8	30	94	18	30	84	28	30	74
9	30	93	19	30	83	29	30	73
10	30	92	20	30	82	30	30	72

Attack:

1. Generate constraints based on data $\leftarrow$ "Easy"

2. Find any microdata consistent w/ constraints

(- This example: 1 feasible sol'n)

"Hard" NP-Hard

"Easy"? SAT Solver, Integer Programming

- Toy example, but can be scaled up
- Tutorial by Abowd
- Attacked 2010 Census
 - Reconstruct microdata for 100% of records
 - ↳ $\exists$ a feasible sol'n
 - Matched 46% of pop. exactly
 - Allow ± 1 year age $\rightarrow$ 71% of pop.
- Commercial databases $\rightarrow$ linkage attack
- Match rows w/ microdata.
- Match, reidentify ≥ 50 million Americans
- Legal liability
- 2020 US Census will use DP

2. Dinur-Nissim Attack [Dwork-Roth '04]

- Database has several "rows" $\leftrightarrow$ points $\leftrightarrow$ individual's data
- Columns $\leftrightarrow$ features $\leftrightarrow$ dimensions
- Most cols are identifiers (public, for now)
- One col is secret

$\hookrightarrow 0$ or 1

- $d \in \{0,1\}^n$ is vector of secret bits

identifiers				sensitive info
Name	Postal Code	Date of Birth	Sex	Has Disease?
Alice	K8V7R6	5/2/1984	F	1
Bob	V5K5J9	2/8/2001	M	0
Charlie	V1C7J	10/10/1954	M	1
David	R4K5T1	4/4/1944	M	0
Eve	G7N8Y3	1/1/1980	F	1

$$d = [1, 0, 1, 0, 1]$$

- Data curator $\rightarrow$ answer queries, while "preserving privacy"
- Analyst $\rightarrow$ ask q's about the dataset

$\hookrightarrow$ Ask: "How many of the rows satisfying COND have "Has Disease? = 1"?"

E.g. COND: "Name = Alice" OR "Name = Charlie" OR "Name = David"

True answer = 2

- Abstractly: can specify subsets of $[n]$

"Subset query"

$S \in \{0,1\}^n$ is query vector.

$S_i = 1$ if i is in the subset we're querying.

$S_i = 0$ else.

$$S = [1, 0, 1, 1, 0], d = [1, 0, 1, 0, 1] \rightarrow A(S) = 2$$

True answer to query S : $A(S) = d \cdot S$

- Curator: Receive a query set S , output response $r(S)$

- Bad idea: Letting $r(S) = A(S)$

$\hookrightarrow$ Let $S = \{1\}$

$r(S) =$ "Noised" version of $A(S)$

Curator output: $r(S)$ $|r(S) - A(S)| \leq \epsilon$ Not assuming $r(S)$ is "random" Curator can "cleverly" perturb $A(S)$

Def: An algo is "blatantly non-private" if an adversary can construct a db $c \in \{0,1\}^n$ s.t. it matches the true db d in all but $o(n)$ entries.

→ Adversary can construct c which matches d in 99.9% of the entries!

"Reconstruction attacks"

→ fairly general schemes are non-private.

Thm: [DN'03]: If the analyst can ask 2^n q's and the curator answers with error E , then the adversary can reconstruct the database in all but $4E$ positions.

- If $E = \frac{n}{401}$, then we can match true db on 99% of entries

- If $E = o(n)$, b.n.p.

Proof:

Attack 1: analyst asks all 2^n queries, $S = [1, 0, 0, \dots]$
 $S = [0, 1, 0, 0, \dots]$
 $S = [1, 1, 0, \dots]$
 $r(S) \forall S$

2. Output any db c , which is "consistent" with answers.

- For each $c \in \{0,1\}^n$, if $\exists$ a query set S , s.t. $|\sum_{i \in S} c_i - r(S)| > E$, rule out c .

- Output any c which is never ruled out.

- Note $|\sum_{i \in S} d_i - r(S)| = |A(S) - r(S)| \leq E$.

d won't be ruled out.

- Need to ensure what we output isn't bad.

$I_0 = \{i \mid d_i = 0\}, I_1 = \{i \mid d_i = 1\}. I_0 \cup I_1 = [n]$

Suppose c is output

Then, by this attack defn, (a) $|\sum_{i \in I_0} c_i - r(I_0)| \leq E$

(c) $|\sum_{i \in I_1} c_i - r(I_1)| \leq E$

Curator's restriction (b) $|\sum_{i \in I_0} d_i - r(I_0)| \leq E$ (a)+(b) $\rightarrow c$ and d differ by $\leq 2E$ entries in I_0
 (d) $|\sum_{i \in I_1} d_i - r(I_1)| \leq E$ (c)+(d) $\rightarrow c$ and d " " I_1

$\Rightarrow c$ and d differ by $\leq 4E$ entries overall. $\square$

Thm [DN'03]: If analyst asks $O(n)$ q 's,
 curator adds $\pm O(\alpha \sqrt{n})$ noise,
 then adv. can reconstruct all but $O(\alpha^2)$
 whp. positions.

(f. 2^n q 's, $E = O(n)$,
 Here: $O(n)$, $E = O(\sqrt{n})$)

Attack: 1. Choose $O(n)$ subsets S , uniformly at random
 2. Output any database consistent w/ results.
 $\hookrightarrow$ Can be done efficiently w/ linear program.

Intuition: Suppose $c, d, S \in \{\pm 1\}^n$ rather than $\{0, 1\}^n$
 Suppose d and some fixed c differ in $\Omega(n)$ coords.
 - A ~~rand~~ query will "witness" this large diff.
 Consider $(c-d) \cdot S = \sum_{i \in [n]} (c_i - d_i) S_i$, where S is ~~var~~ $u.a.r.$ from $\{\pm 1\}^n$

If $c_i = d_i$, term = 0

Else, $(c_i - d_i) S_i$ is ± 1 w.p. $\frac{1}{2}$ each,

$\Rightarrow (c-d) \cdot S \approx \text{Bin}(\Omega(n), \frac{1}{2})$ (scaled, shifted) $\Rightarrow$ anti-concentration, w.p. $\geq \frac{9}{10}$
 $|c-d \cdot S| = \Omega(\sqrt{n})$

If we choose many query sets S ,
at least one will have $|c-d \cdot S| = \Omega(\sqrt{n})$ w.h.p.

$E = O(\sqrt{n})$ more than
censor can
mask

Union bound: Fixed c is eliminated w.h.p.

UB. over $\approx 2^n$ "bad" db gives result.

Thm 8.2 in [DR'14].

Attack 1: 2^n q 's, noise $O(n)$

2: $O(n)$ q 's, noise $O(\sqrt{n})$ - Dwork-McSherry-Talwar '07

$\approx 75\%$ of queries have $E = O(\sqrt{n})$
rest have $E = \infty$

"Tight" by DP.

3. Reconstruction in Practice

- Aircloak challenge

- Diffix

↳ unlimited # of queries on sensitive db

↳ preserving privacy

↳ giving "high accuracy" (outperform)

Effective reconstruction → \$5,000 USD

Similar setting

- analyst unchanged

- curator: Before: ϵ is fixed

Now: ϵ of a query is $\approx \sqrt{\# \text{ of CONDs.}}$

COND = $\left. \begin{array}{l} \text{User} = \text{Alice} \text{ OR} \\ \text{User} = \text{Charlie} \text{ OR} \\ \text{User} = \text{David} \end{array} \right\} 3$

Won't answer: - suppression of small counts
- no ORs in conditions

[DN'03] Chooses rand. queries

1. Choose subset

2. Generate COND to specify subset

Breaks, naively.

"How to generate random queries in Diffix?"

[Cohen-Nissim '20]

- Specify "random" subsets with 1 additional COND.

Their queries:

- each row has a client-id.

↳ have four variables: $\underbrace{\text{mult, exp, d}}_{\text{numbers}}, \underbrace{\text{pred}}_{\text{true/false q.}}$

"Does the d -th digit of $(\text{mult} \times \text{clientid})^{\text{exp}}$ satisfy pred ?"

query $\text{client-id} = 1$
 $(\text{mult} = 17, \text{exp} = 0.5, d = 3, \text{pred} = \text{"is digit even?"})$

$(17.1)^{0.5} \approx 4.1231...$

↑
even? yes.

```
SELECT count(clientId) FROM loans
WHERE floor(100 * ((clientId * 2)^0.7) + 0.5)
      = floor(100 * ((clientId * 2)^0.7))
AND clientId BETWEEN 2000 and 3000
AND loanStatus = 'C'
```


mult=2, exp=0.7, d=2
Pred = "is digit < 5?"

Answered all secret info perfectly.

\$5,000

New Prefix, new challenge

- Census recon:

- Dinur-Nissim $\leftarrow$ Important

- Cohen-Nissim $\leftarrow$ Practical